

ML-Driven DLP Incident Severity Classification (Netskope)

Project Type: Internal Security Automation / Applied Machine Learning

1. Project Objective

Design and implement a supervised machine learning system that consumes Netskope DLP logs, leverages analyst-labeled incidents as ground truth, and automatically predicts incident severity to enable auto-closure and escalation workflows.

2. Scope Definition

In Scope:

- Netskope Email, Web, and Cloud DLP incidents
- Manual analyst labeling
- Severity classification automation

Out of Scope (Phase 1):

- LLM-based analysis
- User behavior analytics
- Cross-vendor correlation

3. Success Criteria

- Model accuracy $\geq 80\%$
- High-severity false negatives $< 5\%$
- Explainable predictions
- $\geq 40\%$ analyst time reduction

4. Architecture Overview

Netskope Logs → Manual Labeling → Feature Engineering → ML Training → Severity Prediction → SOC Action

5. Phase-wise Execution Plan

- Phase 0: Environment Setup (Python, virtualenv, libraries) – Day 1
- Phase 1: Data Ingestion and Log Export – Day 2–3
- Phase 2: Labeling Schema and Guidelines – Day 4
- Phase 3: Manual Incident Labeling (2k–5k incidents) – Day 5–10
- Phase 4: Feature Engineering – Day 11–12
- Phase 5: Model Training (Logistic Regression, Random Forest) – Day 13–14
- Phase 6: Model Evaluation and Validation – Day 15
- Phase 7: Inference Pipeline Development – Day 16
- Phase 8: Automation Logic (Auto-close / Escalate) – Day 17
- Phase 9: Analyst Review UI (Optional) – Day 18–19

- Phase 10: Retraining and Drift Monitoring – Day 20

6. Deliverables

- Labeled Netskope DLP dataset (CSV)
- Labeling guidelines document
- Feature engineering scripts
- Trained ML model and evaluation report
- Inference and automation logic